Maganti Ramesh

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Summary

AI/ML engineer and Computer Science graduate from IIIT Vadodara who gets AI systems into production rather than leaving them in notebooks. Built the AI agents behind a live marketing platform an international client uses daily, fine-tuned an open model past its published benchmark, and interned at NIT Warangal and IIT Jammu. Strongest with LLMs, RAG and agentic systems.

Experience

AI/ML Engineer Intern **Dec 2025 – Jun 2026**
Testaing (Artificial General Intelligence) | SocialJet *Remote*

- Built AI agents that took over work the team was doing by hand: listening to recorded sales calls and drafting the client proposal from them, then writing and negotiating the outreach messages sent to creators.
- Built the engine that picks which creators suit a brand, using vector search to match on both style and budget and turning hours of manual shortlisting into a few seconds. The platform runs live on AWS for an international client.

Research Intern – Optimizing GELU Activation Performance **May 2025 – Aug 2025**
National Institute of Technology, Warangal *Onsite*

- Built CachedGELU, a faster version of GELU, a small maths step that every neural network runs millions of times, by swapping the slow calculation for a quick lookup table.
- It ran **2.23× faster** on a 100-million-value benchmark with almost no loss in precision, and cut CIFAR-10 training time by **35.35 seconds** over 30 epochs at the same accuracy. Also tested inside BERT against other activation functions.

Research Intern – Android Malware Classification **May 2025 – Jul 2025**
Indian Institute of Technology, Jammu *Remote*

- Rebuilt a published Android malware detector that spots harmful apps from how they behave on the phone, sorting them into five types with **92.94% accuracy** on the CICMalDroid 2020 dataset.
- Trained it further using unlabelled apps as well (semi-supervised learning), then measured exactly how much that extra data helped compared with the standard approach.

Projects

Fine-Tuned LLM for Schema-Compliant JSON Output | *GitHub*

- Trained a small open model to reply in the exact data format an application asks for, which is something small models usually get wrong. On the standard public benchmark its accuracy rose from **49% to 67%**, improving on 7 of the 8 test sets and beating the published scores for models of the same size.
- Built the whole pipeline: created the training data, checked every answer automatically while the model trained, and kept the test questions fully separate so the results stayed honest.

AI Disease Diagnosis from X-Rays and CT Scans | *GitHub* | *Live demo*

- Trained three image models to read CT and X-ray scans, reaching **97.29%** accuracy on lung cancer, **99.96%** on kidney disease and **99.20%** on Covid-19.
- Built a web app that highlights the exact area of the scan behind each result, so a doctor can see why the model reached its answer instead of just trusting a number.

Agentic RAG Document Search System | *GitHub*

- Built a document search assistant where the AI decides on its own whether to look inside the uploaded files, search Wikipedia, or simply answer from what it already knows.
- Handled everything behind it, from reading and splitting the documents to the search index and the web app on top, and set it up so the underlying model can be swapped without rewriting code.

Technical Skills

Languages: Python, C, SQL, JavaScript, HTML/CSS
ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, EfficientNet, NumPy, Pandas, Matplotlib
Generative AI: LLMs, Fine-tuning (QLoRA/LoRA, Unsloth), RAG, Agentic AI, LangChain, LangGraph, LangSmith, DSPy, ReAct, MCP, Prompt Engineering, Embeddings, Vector Search
Backend & Cloud: FastAPI, Flask, Streamlit, ReactJS, AWS (DynamoDB, S3, ECS), GCP, Azure, Docker, Kubernetes, CI/CD
Databases: MySQL, SQL Server, MongoDB, Firebase, FAISS and vector stores
MLOps: Model deployment, versioning, monitoring, evaluation (ROC, precision/recall), benchmarking, Git, Linux

Education

Indian Institute of Information Technology Vadodara (IIIT Vadodara) **Nov 2022 – May 2026**
B.Tech in Computer Science and Engineering *India*

Achievements & Certifications

Best Intern of the Month, Testaing (Apr 2026) for shipping production AI systems. | **Kaggle: Top 5%** (189 of 3,800+).
Certifications: NVIDIA Fundamentals of Deep Learning and LLM Customization; Udemy Generative AI; HackerRank SQL (Intermediate).